

WEEK - 1 PROJECT BRIEF

Intern ID: TBI-26100998

App Name: AI-Powered Crop Advisory Chatbot (Prototype)

One-Line Pitch:

An AI-powered chatbot that provides farmers with crop recommendations, disease guidance, and farming advice through natural-language conversations.

Target User:

Small and medium-scale farmers, agricultural students, and rural communities who need quick access to reliable agricultural information. The application helps users make informed decisions regarding crop selection, pest management, irrigation practices, and fertilizer usage, especially in areas where access to agricultural experts may be limited.

Core Features:

- Crop Recommendation – Suggests suitable crops based on season, soil type, and local conditions.
- Disease & Pest Guidance – Provides information about common crop diseases and pest-control methods.
- Farming Q&A Chatbot – Allows users to ask agriculture-related questions and receive instant responses.
- Weather-Based Suggestions – Offers basic farming recommendations using weather and seasonal information.
- Multilingual Support – Enables users to interact with the chatbot in regional languages.

AI Feature:

The Farming Q&A Chatbot uses the OpenAI API to generate intelligent and context-aware responses to agricultural queries. OpenAI is selected because of its strong natural language understanding capabilities and ability to provide conversational assistance.

Tech Stack:

- Frontend: Next.js
- Styling: Tailwind CSS
- Backend: FastAPI
- Database: MongoDB
- Authentication: Clerk Authentication
- Deployment: Vercel (Frontend) + Render (Backend)